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CISC-121
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Assignment 1

↳ Psudocode

1. Cyclic Rotation

```
FUNCTION rotateArray(A, k)
    n = length of A

    IF n == 0 THEN
        return A

    k = k mod n
    result = list of n zeroes

    FOR i from 0 to n - 1
        position = (i + k) mod n
        result[position] = A[i]

    return result
```

2. Perm Check

```
FUNCTION checkPerm(A)
    n = length of A
    seen = list of n + 1 False values

    FOR each x in A
        IF x < 1 OR x > n THEN
            return 0
        IF seen[x] == True THEN
            return 0
        seen[x] = True

    return 1
```

3. Frog Jmp

```
FUNCTION frogJumps (X, Y, D)
    distance = Y - X
    jumps = distance DIV D

    IF distance MOD D != 0 THEN
        jumps = jumps + 1

    return jumps
```

4. Passing Cars

```
FUNCTION countP_Cars (A)
    eastCars = 0
    passing = 0

    FOR each car in A
        IF car = = 0 THEN
            eastCars = eastCars + 1
        ELSE
            passing = passing + eastCars

        IF passing > 1,000,000,000 THEN
            return -1

    return passing
```

5 Missing Integer

```
FUNCTION findMissing (A)
    n = length of A
    seen = list of n+2 False values

    FOR each x in A
        IF x >= 1 AND x <= n THEN
            seen[x] = True

    FOR i from 1 to n+1
        IF seen[i] = = False THEN
            return i
```

```python
#1) CyclicRotation
def rotateArray(A,K):
    n = len(A)
    if n == 0:
        return A
    k = K%n
    result = [0]*n
    for i in range(n):
        pos = (i+k)%n
        result[pos] = A[i]
    return result

#2) PermCheck
def checkPerm(A):
    n = len(A)
    seen = [False]*(n+1)
    for x in A:
        if x<1 or x>n:
            return 0
        if seen[x]:
            return 0
        seen[x] = True
    return 1

#3) FrogJmp
def frogJumps(X,Y,D):
    distance = Y-X
    jumps = distance//D
    if distance%D != 0:
        jumps = jumps + 1
    return jumps

#4) PassingCars
def countP_Cars(A):
    eastCars = 0
    passing = 0
    for car in A:
        if car==0:
            eastCars+=1
        else:
            passing+=eastCars
```

```
42          if passing > 1000000000:
43              return -1
44      return passing
45
46  #5) MissingInteger
47  def findMissing(A):
48      n = len(A)
49      seen =[False]*(n+2)
50      for x in A:
51          if 1⩽x⩽n:
52              seen[x] = True
53      for i in range(1,n+1):
54          if not seen[i]:
55              return i
56
57
58  #test lines
59  #1
60  print(rotateArray([3,8,9,7,6],3))
61  #2
62  print(checkPerm([4,1,3,2]))
63  #3
64  print(frogJumps(10,85,30))
65  #4
66  print(countP_Cars([0,1,0,1,1]))
67  #5
68  print(findMissing([1,3,6,4,1,2]))
69
```

Anthropic(2026, November 8). Pseudocode and code review [Generative AI chat]. Claude. [https://claude.ai/share/17b42f69-5b8a-4537-8ddf-1925174df5c0]